MILITARY SPECIFICATION
SILICA FABRIC, PHENOLIC IMPREGNATED

This specification has been approved by the
Bureau of Naval Weapons, Department of the Navy

1. SCOPE

1.1 Scope. This specification covers one type of chopped-silica fabric impregnated with a phenolic resin containing silica reinforcement, and supplied in approximate 1/2-inch squares.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-P-116	Preservation, Methods of
MIL-R-9299	Resin, Phenolic, Low-Pressure Laminating

STANDARDS

Military

MIL-STD-129	Marking for Shipment and Storage

Department of the Navy
Bureau of Naval Weapons

OS 9349	Fabric, Woven, Vitreous Fiber, For Plastic Laminates

(When requesting any of the above documents, give the title and complete designation of the item shown above. Copies of this specification and other unclassified specifications, standards, and publications required by contractors in connection with specific procurement functions may be obtained from the Commanding Officer, Navy Supply Depot (CDS), 5801 Tabor Avenue, Philadelphia, Pennsylvania, 19120.)

FSC -8305

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T05:08Z
Check the source to verify that this is the current version before use.

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal, shall apply.

PUBLICATIONS

American Society for Testing and Materials

ASTM D 792-60	Method of Test for Specific Gravity and Density of Plastics
ASTM D 695-54	Method of Test for Compressive Properties of Rigid Plastics
ASTM D 651-48	Method of Test for Tensile Strength of Molded Electrical Insulating Materials

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania, 19103.)

3. REQUIREMENTS

3.1 Preproduction sample. Unless otherwise specified in the contract or order, preproduction samples of the fabric shall be manufactured using the methods and procedures proposed for the production. The sample will be tested as specified in Section 4 herein and is for the purpose of determining that, prior to starting production, the contractor's production methods are capable of yielding items that comply with the technical requirements of the contract. After satisfactorily passing all the preproduction tests specified herein, no changes in raw materials and processing of materials for production shall be made without prior written approval of the procuring activity.

3.2 Data requirements. No data is required by this specification or by applicable documents referenced in Section 2 unless specified in the contract or order (see 6.2).

Source: https://assist.dla.mil -- Downloaded: 202640-23T05:08Z
Check the source to verify that this is the current version before use.

3.3 Material. The material shall be composed of a resin meeting Specification MIL-R-9299, Type II, Class 2, a resin reinforcement (minimum 98-percent pure silica) a silica fabric meeting OS 9349, Type III. The reinforcement shall be no less than 5 nor more than 10 percent of the total weight of the impregnated fabric.

3.4 Physical and chemical properties

3.4.1 Uncured impregnated fabric

3.4.1.1 Resin solids content. The resin solids content of the uncured material shall be 25 ± 2 percent.

3.4.1.2 Volatile content. The volatile content of the uncured material shall be 3.75 ± 1.25 percent.

3.4.1.3 Flow. The flow percent shall be 9.5 ± 5.5 percent.

3.4.2 Cured impregnated fabric

3.4.2.1 Specific gravity. The minimum specific gravity of the cured material shall be 1.77.

3.4.2.2 Compressive strength. The minimum compressive strength of the cured material shall be 26,000 pounds per square inch (psi).

3.4.2.3 Tensile strength. The minimum tensile strength of the cured material shall be 8,500 psi.

3.5 Workmanship. The material shall be uniform in quality and shall be free from impurities and other defects that could adversely affect its use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T05:08Z
Check the source to verify that this is the current version before use.

right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. Inspection of the fabric shall be classified as follows:

a. Preproduction inspection (see 4.4)
b. Quality conformance inspection (see 4.5)

4.3 Sampling

4.3.1 Preproduction sample. A preproduction sample of sufficient material shall be used to make two specimens for each of the tests. Each sample, manufactured in accordance with 3.1, shall be subjected to the preproduction tests detailed in 4.4 at an activity designated by the procuring activity. Further production of the impregnated fabric by the supplier, prior to approval of the preproduction sample, shall be at the supplier's risk.

4.3.2 Quality conformance inspection sampling. Unless otherwise specified, sufficient material shall be taken from each lot to perform the tests as specified in 4.5.

4.3.3 Inspection lot. An inspection lot of the impregnated silica fabric shall consist of all material presented for acceptance at one time and produced in a single manufacturing batch under homogeneous conditions of manufacture.

4.4 Preproduction inspection. The preproduction sample shall satisfactorily pass the quality conformance inspections detailed in 4.5. In addition, the following test shall be performed.

4.4.1 Silica purity test. This test shall be run prior to the material being added to the resin. Weigh a platinum crucible with cover to the nearest 0.1 milligram (mg) (weight A). Place about 0.50-gram sample into the crucible, cover, and ignite for 10 minutes under the full flame of a

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T05:08Z
Check the source to verify that this is the current version before use.

Meeker burner. After ignition, cool in a desiccator and weigh with cover on to the nearest 0.1 mg (weight B). Add 5 drops of 1:1 sulfuric acid and sufficient hydrofluoric acid (47-percent reagent grade) to fill the crucible to about 50 percent capacity. Place crucible in sand bath and evaporate slowly (to prevent spattering) under a hood until fuming ceases. Again add 5 drops of 1:1 sulfuric acid and sufficient hydrofluoric acid (47 percent reagent grade) to fill the crucible to about 50-percent capacity. Place crucible in sand bath and evaporate slowly (to prevent spattering) under a hood until fuming ceases. After the second evaporation, ignite crucible for 10 minutes under the full flame of the Meeker burner. Cool in desiccator and weigh with cover on to the nearest 0.1 mg (weight C).

Calculations:

$$\text{Percent silica } (SiO_2) = \frac{\text{Weight B - Weight C}}{\text{Weight B - Weight A}} \times 100$$

4.4.2 Specific gravity test. From each sample, mold test specimens of the impregnated fabric at 2000 ± 100 psi and 310 ± 5 degrees Fahrenheit (F) for 6 minutes. Determine the specific gravity in accordance with ASTM D 792-60.

4.4.3 Compressive strength test. From each sample, mold test specimens of the impregnated fabric at 6000 ± 100 psi and 310 ± 5 degrees F for 10 minutes. Determine the compressive strength in accordance with ASTM D 695-54.

4.4.4 Tensile strength test. From each sample, mold test specimens of the impregnated fabric at 4000 ± 100 psi and 310 ± 5 degrees F for 6 minutes. Determine the tensile strength in accordance with ASTM D 651-48.

4.5 Quality conformance inspection

4.5.1 Visual examination. Visually examine all containers in the lot to determine compliance with 3.5 and Section 5.

Source: https://assist.dla.mil -- Downloaded: 202610-23T05:08Z
Check the source to verify that this is the currentversion before use.

4.5.2 Volatiles and resin solids content test. The volatiles and resin solids content of the uncured impregnated fabric shall be determined by weighing 5 to 6 grams of sample to the nearest milligram (0.001 mg) into a previously fired (1450 degrees F) and tared crucible. Designate this weight as W_1. Condition the sample and crucible in a circulating-air oven at 325 ± 5 degrees F for 20 minutes; remove, cool in a desiccator and reweigh to obtain the sample weight. Designate this weight as W_2. Using a tripod, clay triangle, and Bunsen burner, carefully heat this same sample and crucible to eliminate combustible material and carbonize resin. When carbonization is complete, place the sample and crucible in the muffle furnace. Condition at 1450 ± 10 degrees F for 3 hours; remove, cool in desiccator and reweigh to obtain the final weight. Designate this weight as W_3.

Calculations:

A. Volatile content at given conditions

$$\text{Percent volatiles} = \frac{W_1 - W_2}{W_1} \times 100$$

where:

W_1 = Original sample weight

W_2 = Sample weight after 20 minutes at 325 ± 5 degrees F

B. Resin solids content

$$\text{Percent resin solids} = \frac{W_2 - W_3}{W_2} \times 100$$

where:

W_2 = Sample weight after 20 minutes at 325 ± 5 degrees F

W_3 = Sample weight after 20 minutes at 325 ± 5 degrees F and and 3 hours at 1450 ± 10 degrees F

4.5.3 Flow test. All material shall be brought to room temperature

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T05:08Z
Check the source to verify that this is the current version before use.

before starting the flow test. Weigh a sample of approximately 50 grams to the nearest 0.01 gram. Designate this as W_1. The sample shall be preformed in a 2-3/4 ± 1/16 inch diameter mold under 5000 ± 25 psi pressure. Allow the preform to dwell 2-3 seconds. Place the sample into mold and close under a pressure of 2000 ± 25 psi at a temperature of 300 ± 5 degrees F for 5-6 minutes. The closing time shall be 6-7 seconds. Take the sample from the mold and remove the sprue carefully and cut off the first measurable 1.00 inch adjoining the disc. Weigh the disc to the nearest 0.01 gram and designate this as W_2.

$$\text{Percent flow} = \frac{W_1 - W_2}{W_1} \times 100$$

4.6 Acceptance criteria

4.6.1 Preproduction. Failure of any test to meet the requirements of this specification shall be cause for rejection of the lot.

4.6.2 Quality conformance. Failure of any sample to meet any requirement of this specification shall be cause for rejection of the lot.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging and packing. Unless otherwise specified in the contract or order, minimum requirements for preservation, packaging and packing shall be in accordance with MIL-P-116, Method III.

5.2 Marking. Each container shall be marked in accordance with MIL-STD-129. Marking shall include, but not be limited to, the following information:

a. Manufacturer's name and location
b. Material trade name
c. Net weight or volume
d. Lot number, batch number and date of manufacture
e. Shelf life or storage limitations
f. Number and revision letter of this specification

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T05:08Z
Check the source to verify that this is the current version before use.

6. NOTES

6.1 Intended use. The material purchased in accordance with this specification is intended to be used as a component of rocket motors.

6.2 Ordering data. Procurement documents should specify, but not be limited to, the following information:

a. Title, number and revision letter of this specification
b. Minimum lot size, if applicable
c. Whether preproduction sample is required
d. Place of delivery
e. Size of container
f. Request for test data

Source: https://assist.dla.mil -- Downloaded: 202640-23T05:08Z
Check the source to verify that this is the currentversion before use.

FOLD

DEPARTMENT OF THE NAVY
Bureau of Naval Weapons
Washington, D. C. 20360

POSTAGE AND FEES PAID
NAVY DEPARTMENT

OFFICIAL BUSINESS

BUREAU OF NAVAL WEAPONS (RREN-5)
DEPARTMENT OF THE NAVY
WASHINGTON, D. C. 20360

FOLD

Source: https://assist.dla.mil -- Downloaded: 202610-23T05:08Z
Check the source to verify that this is the current version before use.

SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION
MIL-S-81256(WP) SILICA FABRIC, PHENOLIC IMPREGNATED

ORGANIZATION *(Of submitter)* | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

DD FORM 1 APR 83 1426

C-8279

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T05:08Z
Check the source to verify that this is the current version before use.